

The expected characteristic polynomial of a random lift of a
cycle:
an elementary proof, a generating function,
and a machine-checked formalization

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Abstract

Let $\Phi_{n,r}(x)$ denote the expected characteristic polynomial of a uniformly random permutation r -lift of the cycle C_n . We give a short, self-contained derivation of the closed form $\Phi_{n,r}(x) = \chi_{C_n}(x) \cdot U_{r-1}(T_n(x/2))$, where T, U are Chebyshev polynomials and $\chi_{C_n}(x) = 2T_n(x/2) - 2$; equivalently, the d -matching polynomial of C_n (Hall–Puder–Sawin) is $U_d(T_n(x/2))$. This statement is equivalent, via the Hall–Puder–Sawin identity, to a conjecture of Hall proved combinatorially by Cochran, Groothuis, Herring, Rohatgi and Stucky (2018); our proof is independent, passing through the exponential formula and a Chebyshev generating function, and yields as byproducts the generating function $\sum_{d \geq 0} \mu_{d,C_n}(x) z^d = (1 - 2T_n(x/2)z + z^2)^{-1}$, a Fibonacci–Lucas evaluation, and a parity theorem for the quotient. The entire derivation from cycle weights to the closed form — including the exponential formula for S_r — is formalized in Lean 4 over Mathlib, with no axioms beyond Lean’s standard three. The same argument applies to every connected graph of first Betti number one, where it gives $\mu_{d,G} = V_d$ for the three-term recurrence $V_d = \mu_G V_{d-1} - \mu_{G-V(C)}^2 V_{d-2}$ in the ordinary matching polynomials of G and of the forest obtained by deleting the cycle; this answers Question 6.5 of Hall–Puder–Sawin for that family, and we show the Betti-number-one hypothesis cannot be relaxed. For the associated gap problem we prove that every nonzero root of the matching polynomial of a subdivision $S(H)$ of a D -regular graph has absolute value at least $\sqrt{D-1} - 1$, which settles the case $\min(d, r) = 2$ of Problem 1 of Song, Fan and Miao. After the change of variable that centres the band, the resulting polynomial g has subleading coefficient exactly $p(b-2)$; so g is a matching polynomial precisely when $\min(a, b) = 2$, and for $\min(a, b) \geq 3$ its roots are displaced by $b-2$ towards the lower band edge, which is the end that remains open. Both statements are formalized.

1 Introduction and statement

The expected characteristic polynomial of a random lift is the object at the heart of the interlacing-families method of Marcus–Spielman–Srivastava [16], who used the $r = 2$ (signing) case to prove that bipartite Ramanujan graphs exist in every degree; Hall–Puder–Sawin [2] extended the framework to arbitrary coverings. We evaluate it in closed form for the cycle.

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A *permutation r -lift* of a graph G replaces each vertex by r copies and each edge $\{u, v\}$ by the perfect matching $u_i \sim v_{\sigma(i)}$ of a permutation $\sigma \in S_r$ chosen independently and uniformly per edge. For the cycle C_n , gauge-fixing a spanning path reduces the lift to a single permutation $\sigma \in S_r$ on the closing edge, and the lift decomposes into disjoint cycles: each ℓ -cycle of σ contributes a cycle $C_{n\ell}$ (fixed points contribute copies of C_n). Since $\chi_{C_m}(x) = \det(xI - A_{C_m}) = 2T_m(x/2) - 2$, the expected characteristic polynomial is the expected cycle weight

$$\Phi_{n,r}(x) = \mathbb{E}_{\sigma \in S_r} \prod_{\ell \in \text{cycles}(\sigma)} (2T_{n\ell}(x/2) - 2). \quad (1)$$

Theorem 1. *For all $n \geq 1$, $r \geq 1$, with $Y = T_n(x/2)$,*

$$\Phi_{n,r}(x) = U_r(Y) - 2U_{r-1}(Y) + U_{r-2}(Y) = \underbrace{(2T_n(x/2) - 2)}_{\chi_{C_n}(x)} \cdot \underbrace{U_{r-1}(T_n(x/2))}_{\Psi_r(x)}.$$

The factored form identifies the quotient Ψ_r — by the theorem of Hall–Puder–Sawin [2] equating the expected new-eigenvalue characteristic polynomial with the d -matching polynomial ($d = r - 1$) — as

$$\mu_{d,C_n}(x) = U_d(T_n(x/2)),$$

which is equivalent (through $\mathcal{P}_d(2t) = U_d(t)$, $\mathcal{C}_n(2t) = 2T_n(t)$ and the composition identity $U_{dn+n-1} = U_d(T_n)U_{n-1}$) to *Hall's conjecture* $\mathcal{C}_{n,d} = \mathcal{P}_d(\mathcal{C}_n)$, proved in [3]. Our route is different and elementary; we make no use of matching-polynomial combinatorics.

Corollary 2. $\sum_{d \geq 0} \mu_{d,C_n}(x) z^d = \frac{1}{1 - 2T_n(x/2)z + z^2}, \quad \text{and} \quad \sum_{r \geq 0} \Phi_{n,r}(x) z^r = \frac{(1 - z)^2}{1 - 2T_n(x/2)z + z^2}.$

Corollary 3 (real-rootedness). *All roots of $\Phi_{n,r}$ lie in $[-2, 2]$: with $x = 2 \cos \theta$, χ_{C_n} vanishes at $\theta = 2\pi m/n$ and $\Psi_r = \sin(rn\theta)/\sin(n\theta)$ vanishes at $\theta = m\pi/(rn)$ with $r \nmid m$ (when $r \mid m$ the denominator vanishes too and $\Psi_r = \pm r \neq 0$); this leaves exactly $n(r-1)$ roots, matching $\deg \Psi_r$.*

Corollary 4 (Fibonacci–Lucas evaluation). $\Phi_{n,r}(3) = (L_{2n} - 2) F_{2nr}/F_{2n}$, since $T_n(3/2) = L_{2n}/2$ and $U_{r-1}(L_{2n}/2) = F_{2nr}/F_{2n}$.

Corollary 5 (parity). $\Psi_r(-x) = (-1)^{n(r-1)} \Psi_r(x)$: *the quotient is supported on exponents of a single parity, like a matching polynomial. (For bipartite C_n it is an even polynomial for every r ; at $r = 2$ it is the classical matching polynomial $2T_n(x/2)$ of C_n , consistent with Godsil–Gutman [1].)*

2 Proof

Write $f(\ell) = 2T_\ell(Y) - 2$ with $Y = T_n(x/2)$; by the composition identity $T_{n\ell} = T_\ell \circ T_n$ this is the per-cycle factor in (1).

Step 1 (exponential formula). For any commutative ring R and $f : \mathbb{N} \rightarrow R$, the S_r -average $\Phi_r = \mathbb{E}_\sigma \prod_{\ell \in \text{cycles}(\sigma)} f(\ell)$ satisfies

$$r \Phi_r = \sum_{k=1}^r f(k) \Phi_{r-k}, \quad \Phi_0 = 1, \quad (2)$$

the coefficient form of $\sum_r \Phi_r z^r = \exp(\sum_\ell f(\ell) z^\ell / \ell)$. (Proof: group σ by cycle type λ with Cauchy's count $r! / z_\lambda$, mark a part, and re-index; see the Lean file `Paper2ExpFormula.lean` for the fully detailed version.)

Step 2 (Chebyshev generating function). With $f(\ell) = 2T_\ell(Y) - 2$,

$$\sum_{\ell \geq 1} f(\ell) \frac{z^\ell}{\ell} = -\log(1 - 2Yz + z^2) + 2\log(1 - z) \implies \sum_{r \geq 0} \Phi_r z^r = \frac{(1 - z)^2}{1 - 2Yz + z^2}.$$

Extracting coefficients against $\sum U_r(Y) z^r = (1 - 2Yz + z^2)^{-1}$ gives $\Phi_r = U_r - 2U_{r-1} + U_{r-2}$, and the three-term recurrence $U_r + U_{r-2} = 2YU_{r-1}$ collapses this to $(2Y - 2)U_{r-1}(Y)$. $\square$

In the formalization the analytic Step 2 is replaced by an equivalent finite argument: the closed form is shown to satisfy (2) directly (a second-difference induction), and (2) with $\Phi_0 = 1$ determines Φ over a characteristic-zero domain.

3 Graphs of first Betti number one

The proof above uses cycles in exactly one place: gauge-fixing a spanning path leaves a *single* permutation, because C_n has $|E| - |V| + 1 = 1$. The argument therefore applies verbatim to any connected G with $b_1(G) = |E| - |V| + 1 = 1$, a cycle with trees attached, and we record what it gives.

By [2] the expected characteristic polynomial of a random r -lift of *any* graph factors as $\chi_G \cdot \mu_{r-1,G}$; the content below is therefore not that factorization but a closed form for the factor $\mu_{d,G}$ itself. Question 6.5 of [2] asks which parts of the theory of the matching polynomial $\mu_{1,G}$ generalize to $\mu_{d,G}$; the following answers it for $b_1(G) = 1$.

Let C be the unique cycle of G and let $G - V(C)$ be the forest obtained by deleting its vertices. Write μ_H for the ordinary matching polynomial of a graph H , with $\mu_\emptyset = 1$. Define

$$V_0 = 1, \quad V_1 = \mu_G, \quad V_d = \mu_G V_{d-1} - \mu_{G-V(C)}^2 V_{d-2}. \quad (3)$$

Theorem 6. *For every connected G with $b_1(G) = 1$ and every $d \geq 0$, $\mu_{d,G} = V_d$.*

For $G = C_n$ the deleted forest is empty, so $\mu_{G-V(C)} = 1$ and (3) is the recurrence for $U_d(\mu_{C_n}/2) = U_d(T_n(x/2))$: Theorem 1 is the case $\mu_{G-V(C)} = 1$. In general the attached trees enter only through $\mu_{G-V(C)}$; for the triangle with one pendant edge, $\mu_G = x^4 - 4x^2 + 1$ and $\mu_{G-V(C)} = x$.

The proof goes through the twisted characteristic polynomial. Let $e = uv$ be the unique non-tree edge, let $A_G(z)$ be the adjacency matrix of G with the (u, v) entry replaced by z and the (v, u) entry by z^{-1} , and set $F(x, z) = \det(xI - A_G(z))$.

Lemma 7. $F(x, z) = \mu_G(x) - \mu_{G-V(C)}(x)(z + z^{-1})$.

Proof. By the Sachs coefficient theorem $\det(xI - A)$ expands over elementary subgraphs, that is disjoint unions of edges and cycles. Since a permutation uses each of the two twisted entries at most once and $A_G(z)^\top = A_G(z^{-1})$, the determinant is of the form $P(x) + Q(x)(z + z^{-1})$. Replacing z by ± 1 and averaging kills exactly the elementary subgraphs whose cycle traverses e ; as G has the single cycle C , those are the subgraphs containing C , so $P = \mu_G$, and the surviving terms contribute $Q = -\mu_{G-V(C)}$, a cycle component carrying the Sachs weight -2 . $\square$

Proof of Theorem 6. Gauge-fix a spanning tree: the r -lift is determined by one permutation $\sigma \in S_r$ on e , and its components are indexed by the cycles of σ , the component of an ℓ -cycle being the ℓ -fold cyclic cover G_ℓ . Decomposing the $\mathbb{Z}/\ell$ action into characters gives the classical factorization $\chi_{G_\ell}(x) = \prod_{j=0}^{\ell-1} F(x, \omega^j)$ with $\omega = e^{2\pi i/\ell}$ [14, 15]. With P, Q as in Lemma 7 and $y = -P/(2Q)$,

$$\chi_{G_\ell} = \prod_{j=0}^{\ell-1} \left(P + 2Q \cos \frac{2\pi j}{\ell} \right) = (-Q)^\ell (2T_\ell(y) - 2),$$

so by the exponential formula (2), with $w = -Qz$,

$$\sum_{r \geq 0} \Phi_{G,r} z^r = \exp \left(\sum_{\ell \geq 1} (2T_\ell(y) - 2) \frac{w^\ell}{\ell} \right) = \frac{(1-w)^2}{1-2yw+w^2},$$

and extracting coefficients as in Section 2 gives $\Phi_{G,r} = \chi_G \cdot (-Q)^{r-1} U_{r-1}(y)$. Comparing with $\Phi_{G,r} = \chi_G \cdot \mu_{r-1,G}$ [2] yields $\mu_{d,G} = (-Q)^d U_d(y)$, which satisfies (3). $\square$

Since $U_d(y) = 2^d \prod_{k=1}^d (y - \cos \frac{k\pi}{d+1})$, the theorem can be restated as a product, $\mu_{d,G}(x) = \prod_{k=1}^d F(x, e^{ik\pi/(d+1)})$, whence:

Corollary 8. *For $b_1(G) = 1$, $\mu_{d,G}$ is a product of d real-rooted polynomials, and its roots are exactly the eigenvalues of the Hermitian matrices $A_G(e^{ik\pi/(d+1)})$, $1 \leq k \leq d$.*

This is the real-rootedness theorem of [2] for this family, with the roots identified. The localization itself — that a vanishing V_d forces $|\mu_G| \leq 2|\mu_{G-V(C)}|$, so that a root can never sit in a spectral gap — is machine-checked in Lean 4 as `cV_root_mem_band`, resting on `cU_ne_zero_of_one_le_abs`: the Chebyshev polynomial U_d has no root of absolute value at least one.

The roots can be located much more precisely than that. Since $b_1(G) = 1$ the group $\pi_1(G)$ is $\mathbb{Z}$, so the universal cover T of G is the $\mathbb{Z}$ -cover, and T is a periodic decorated chain whose adjacency operator decomposes by Floquet–Bloch into the fibres $A_G(e^{i\theta})$. Hence

$$\text{spec}(T) = \bigcup_{\theta} \text{spec } A_G(e^{i\theta}) = \left\{ x : \exists s \in [-1, 1], \mu_G(x) = 2s \mu_{G-V(C)}(x) \right\},$$

a finite union of bands, together with a flat band at each common zero of μ_G and $\mu_{G-V(C)}$, where $F(x, \cdot)$ vanishes identically.

Corollary 9. *For $b_1(G) = 1$ the roots of $\mu_{d,G}$ are exactly the solutions of $\mu_G(x) = 2 \cos(\frac{k\pi}{d+1}) \mu_{G-V(C)}(x)$ for $1 \leq k \leq d$. In particular every root of every $\mu_{d,G}$ lies in $\text{spec}(T)$, and as $d \rightarrow \infty$ the roots equidistribute on $\text{spec}(T)$ with respect to its density of states.*

Proof. Immediate from Theorem 6 in the form $\mu_{d,G} = \prod_k F(x, e^{ik\pi/(d+1)})$ and $F(x, e^{i\theta}) = \mu_G - 2 \cos \theta \mu_{G-V(C)}$. The equidistribution is the arcsine law for $\cos(k\pi/(d+1))$ transported through the fibre map. $\square$

Hall–Puder–Sawin place the roots of $\mu_{d,G}$ in the Ramanujan *interval* $[-\rho, \rho]$, ρ being the spectral radius of T . Corollary 9 places them in $\text{spec}(T)$, which is a union of bands and is a proper subset of that interval whenever T has a spectral gap: no root of any $\mu_{d,G}$ can ever enter a gap. The two statements coincide exactly when the band structure is trivial, and that is precisely the case of a cycle, where $\mu_{G-V(C)} = 1$ and $\text{spec}(T) = [-2, 2]$ is the whole interval. Attaching even a single pendant edge opens gaps, and they are not small (computed band structures, proportion of $[-\rho, \rho]$ that is a gap):

G	ρ	gap proportion
C_n	2	0%
triangle + pendant	2.170	36.5%
C_4 + pendant	2.136	45.9%
triangle + P_2	2.214	51.3%
C_4 + P_3	2.189	54.1%
triangle + $K_{1,3}$	2.514	65.9%

So the cycle is not merely the easiest case of Theorem 6: it is the degenerate one, the unique member of the family whose spectrum has no gaps and for which the Ramanujan interval is attained. Any cycle with a tree attached exhibits a band structure that no cycle can display.

A conjecture for all graphs. Corollary 9 is proved by Floquet theory, which is available only because $\pi_1(G) = \mathbb{Z}$. Its *statement*, however, makes sense for every graph, and we know one case of it: at $b_1(G) = 1$ it is Corollary 9, for every d . We therefore propose:

Conjecture 10. *For every finite graph G and every $d \geq 1$, all roots of $\mu_{d,G}$ lie in $\text{spec}(T)$, the spectrum of the universal cover of G .*

The biregular case of Conjecture 10 at $d = 1$ is not new. It is Problem 1 of Song, Fan and Miao [5], posed in 2023: they ask whether the matching polynomial of the incidence graph B_H of a (d, r) -regular hypergraph has no root μ with $0 < |\mu| < |\sqrt{d-1} - \sqrt{r-1}|$, and every (d, r) -biregular bipartite graph is such an incidence graph. Their Theorem 1.8 attaches a consequence: an affirmative answer would give every (d, r) -regular hypergraph a left-sided Ramanujan 2-covering. The only bound in that direction we know of is their Lemma 5.5, which goes the other way, $0 < \mu_\tau(B_H) \leq \max\{\sqrt{d}, \sqrt{r}\}$. We are not aware of any proof, partial result or counterexample. Conjecture 10 extends their problem in two directions, to arbitrary graphs and to all d .

Question 6.3 of [2] asks the corresponding question for graph eigenvalues, and leaves it open: it distinguishes graphs whose nontrivial spectrum lies in the *interval* $[-\rho, \rho]$ from those whose nontrivial spectrum lies in $\text{spec}(T)$, observes that the two notions separate exactly when T has a gap, and gives the biregular tree as the example. Conjecture 10 is the matching-polynomial form of that distinction.

The conjecture strengthens the root-localization theorem of [2], which places the roots in the interval $[-\rho, \rho]$ with ρ the spectral radius of T ; the conjecture places them in $\text{spec}(T)$ itself. The two agree whenever T has no spectral gap — in particular for regular G , where T is a regular tree and $\text{spec}(T) = [-2\sqrt{q}, 2\sqrt{q}]$ is an interval. They differ exactly when T has a gap, and the smallest examples are biregular: for $K_{2,3}$ the universal cover is the $(3, 2)$ -biregular tree, with spectrum $\{0\} \cup \pm[\sqrt{2}-1, \sqrt{2}+1]$ and hence a gap $(0, \sqrt{2}-1)$ inside

the Ramanujan interval. We have tested the conjecture on the biregular family, where the universal cover is an (a, b) -biregular tree with spectrum $\{0\} \cup \pm[|s - t|, s + t]$, $s = \sqrt{a - 1}$, $t = \sqrt{b - 1}$, so the gap is $(0, |s - t|)$. In each case $\mu_{d,G}$ was computed by averaging matching polynomials over all d -covers. No root ever lies in the gap; we also record the smallest nonzero $|\text{root}|$ as a multiple of the gap edge, which measures how close the bound comes to being attained.

G	b_1	type	gap edge	d	min $ \text{root} /\text{gap edge}$
$K_{2,3}$	2	(3, 2)	0.4142	1, 2, 3, 4	2.72, 2.19, 1.94, 1.78
$K_{2,4}$	3	(4, 2)	0.7321	1, 2, 3	1.93, 1.65, 1.52
$K_{2,5}$	4	(5, 2)	1.0000	1, 2, 3	1.66, 1.46, 1.37
$K_{3,4}$	6	(4, 3)	0.3178	1, 2	3.04, 2.31
$K_{3,5}$	8	(5, 3)	0.5858	1, 2	2.10, 1.73
subdivision of K_4	3	(3, 2)	0.4142	1, 2	1.97, 1.67

All of these have $b_1 \geq 2$, so none is covered by Corollary 9. It is worth recording that even $d = 1$ is not formal. There $\mu_{1,G}$ is the ordinary matching polynomial, whose roots are eigenvalues of the path tree of G (Godsil), and the path tree embeds in T ; but eigenvalues of a finite subtree of T need not lie in $\text{spec}(T)$ when T has a gap. For the $(3, 2)$ -biregular tree, whose gap is $(0, \sqrt{2} - 1)$, the path P_8 is a subtree with eigenvalue $2 \cos(4\pi/9) = 0.3473$ inside that gap. So the $d = 1$ case does not follow from the path-tree argument. It is nevertheless a theorem for an infinite family:

Theorem 11. *Let H be D -regular, $D \geq 2$, and let $S(H)$ be its subdivision, so that the universal cover of $S(H)$ is the $(D, 2)$ -biregular tree with spectral gap $(0, \sqrt{D - 1} - 1)$. Then every nonzero root of $\mu_{S(H)}$ has absolute value at least $\sqrt{D - 1} - 1$; that is, Conjecture 10 holds for $S(H)$ at $d = 1$, and the bound is sharp.*

Theorem 11 settles the case $\min(d, r) = 2$ of Problem 1 of [5]. Their uniformity condition allows $r \geq 2$, and they observe that when H is a simple graph the incidence graph B_H is exactly the subdivision $S(H)$; so their question at $r = 2$ asks precisely whether $\mu_{S(H)}$ has no root μ with $0 < |\mu| < |\sqrt{D - 1} - 1|$. By their Theorem 1.8 this gives every D -regular graph, viewed as a $(D, 2)$ -regular hypergraph, a left-sided Ramanujan 2-covering. The case $d = 2$ follows as well, since B_H is a bipartite graph and μ_{B_H} does not distinguish its two sides: a $(2, r)$ -regular hypergraph has for incidence graph the subdivision of an r -regular graph, and the quantity $|\sqrt{d - 1} - \sqrt{r - 1}|$ is symmetric in d and r . The case $\min(d, r) \geq 3$ of their problem remains open, as does Conjecture 10 for $\min(a, b) \geq 3$.

Proof. By the subdivision identity $\mu_{S(H)}(x) = x^{|E| - |V|} \mu_H(x^2 - D)$ [4], the nonzero roots of $\mu_{S(H)}$ are the x with $x^2 = D + \theta$ for θ a root of μ_H . Heilmann–Lieb gives $\theta \geq -2\sqrt{D - 1}$, so with $s = \sqrt{D - 1}$ we get $x^2 \geq s^2 + 1 - 2s = (s - 1)^2$, whence $|x| \geq s - 1$. Sharpness: for cubic H of large girth $\theta_{\min} \rightarrow -2\sqrt{2}$, so the bound is approached. $\square$

The inequality step is machine-checked in Lean 4 as `subdivision_gap`. The theorem is exactly Heilmann–Lieb for H transported through the subdivision identity, and the numbers match: for $H = K_4$, $\mu_{K_4} = x^4 - 6x^2 + 3$ gives smallest positive root $\sqrt{3 - \sqrt{3 + \sqrt{6}}} = 0.8158$, against the gap edge 0.4142; the subdivisions of $K_{3,3}$, the prism, Q_3 , the Wagner, Petersen,

Heawood and Möbius–Kantor graphs give 0.701, 0.714, 0.650, 0.647, 0.607, 0.563, 0.551, decreasing toward the gap edge with the girth and never reaching it.

For $K_{2,3}$ the computation was pushed to $d = 6$ using $\mu_{d,G} = \mathbb{E}[\chi_H]/\chi_G$ over $(d+1)$ -covers, which is polynomial time per cover, together with the reduction of the first free permutation to one representative per cycle type; the ratios continue 1.68 at $d = 5$ and no root in the gap at $d = 6$. That last case is a caution worth recording: in floating point it appeared to produce roots at ± 0.055 , well inside the gap, which an exact integer recomputation showed to be an artefact of evaluating a degree-30 polynomial built from 35×35 characteristic polynomials in double precision.

The last column decreases monotonically in d in every case, which is what the conjecture predicts: by Corollary 9 in the $b_1 = 1$ case the roots equidistribute on $\text{spec}(T)$, so they should approach the gap edge without crossing it. The evidence is therefore not merely that the bound holds but that it is approached, though no computation here reaches the regime where the ratio is close to 1.

The biregular case is one interval, and we have one end of it. Let G be (a, b) -biregular bipartite, $s = \sqrt{a-1}$, $t = \sqrt{b-1}$. Since G is bipartite, $\mu_G(x) = x^\varepsilon f(x^2)$ for a polynomial f and $\varepsilon \in \{0, 1, 2, \dots\}$. Put

$$c = (a-1) + (b-1), \quad m = (a-1)(b-1), \quad g(z) = f(z+c).$$

The universal cover has spectrum $\{0\} \cup \pm[|s-t|, s+t]$, and $(s \pm t)^2 = c \pm 2\sqrt{m}$, so

$$x \in \text{spec}(T) \setminus \{0\} \iff x^2 - c \in [-2\sqrt{m}, 2\sqrt{m}].$$

The change of variable therefore turns Conjecture 10, for biregular G , into a single clean statement:

$$\text{every root of } g \text{ lies in } [-2\sqrt{m}, 2\sqrt{m}], \tag{4}$$

which is a Heilmann–Lieb interval of effective degree $m+1$. The upper end of (4) is known: the path-tree argument of Godsil bounds the roots of μ_G by the spectral radius of the path tree, which for an (a, b) -biregular graph with $a, b \geq 2$ is at most $s+t$ [9, 10]. It is worth noting that [8] proved the corresponding statement only in the regular case; the irregular case is due to Godsil and to [16]. The lower end is the conjecture. *They are the two ends of one interval*, so what is missing is exactly half of a two-sided Heilmann–Lieb theorem for biregular bipartite graphs.

This also explains the boundary between what is proved and what is not, and the explanation is exact.

Theorem 12. *Let G be (a, b) -biregular bipartite with parts of sizes $p \leq q$, degree a on the p -side and b on the q -side, and let $g(z) = f_G(z+c)$ with $c = (a-1) + (b-1)$. Then*

$$[z^{p-1}]g = p(b-2).$$

Consequently the roots of g have mean $-(b-2)$, and g is the matching polynomial of a graph if and only if $b = 2$.

Proof. Counting edges by their smaller side gives $pa = qb$, so $p \leq q$ forces $a \geq b$ and $b = \min(a, b)$. The polynomial f_G is monic of degree p , and its next coefficient is $-m_1$, where m_1 is the number of 1-matchings, that is $|E| = pa$. Hence

$$[z^{p-1}] f_G(z + c) = \binom{p}{1} c - pa = p(a + b - 2) - pa = p(b - 2).$$

The mean of the roots is $-[z^{p-1}]g/p = -(b - 2)$. A matching polynomial of degree p involves only every other power of z , so its coefficient at z^{p-1} vanishes; thus $b = 2$ is necessary. It is sufficient: for $b = 2$ the graph is $S(H)$ for an a -regular H and $g = \mu_H$ by [6]. $\square$

So for $b = 2$ the polynomial g is a matching polynomial, Heilmann–Lieb applies verbatim, and one recovers Theorem 11 and the subdivision identity as the case $c = (a - 1) + 1$ of the change of variable. For $b \geq 3$ it provably is not, and Theorem 12 says more than that it fails: it locates the failure. The roots of g are not centred at the band centre but sit a distance $b - 2$ below it, towards the lower band edge — which is exactly the end of (4) that is open. The asymmetry of the problem and the departure from the matching-polynomial class are the same phenomenon, measured by the single quantity $b - 2$. Examples:

$$g_{K_{3,4}} = z^3 + 3z^2 - 9z - 19, \quad g_{K_{3,5}} = z^3 + 3z^2 - 12z - 24, \quad g_{K_{4,5}} = z^4 + 8z^3 - 6z^2 - 128z - 139,$$

with subleading coefficients $3 = 3(3 - 2)$, $3 = 3(3 - 2)$ and $8 = 4(4 - 2)$ as the theorem requires. Their roots do satisfy (4), with maxima 4.06, 4.48, 6.26 against bounds 4.90, 5.66, 6.93. That class admits a clean description. Since a k -matching leaves $p - k$ vertices of A unmatched,

$$f_G(y) = \sum_k (-1)^k m_k y^{p-k} = \sum_M (-1)^{|M|} \prod_{v \in A \setminus V(M)} y = \mathcal{M}(G; x_A = y, x_B = 1), \quad (5)$$

the multivariate matching polynomial of Heilmann and Lieb under the asymmetric specialization that puts the variable on one side of the bipartition and 1 on the other. So the g are not exotic: they are shifts of matching polynomials in the multivariate sense. This is a reformulation and not an advance: for bipartite G every edge of a matching covers one vertex of each side, so $\mathcal{M}(G; x_A = y, x_B = z)$ depends only on the product yz , and the asymmetric specialization therefore carries exactly the information of the univariate matching polynomial. In particular no quantitative Heilmann–Lieb theorem for non-uniform specializations can help here, and none is available: the multivariate bounds of [8] apply a single graph-wide threshold to every variable.

Deleting a vertex now gives two different recursions, because only A -vertices carry the variable:

$$v \in A: \quad f_G = y f_{G-v} - \sum_{u \sim v} f_{G-v-u}, \quad u \in B: \quad f_G = f_{G-u} - \sum_{v \sim u} f_{G-u-v},$$

whence the cavity recursion $R_A = y - \sum 1/R_B$, $R_B = 1 - \sum 1/R_A$. On the (a, b) -biregular tree, with $P = a - 1$, $Q = b - 1$, its fixed point satisfies $R^2 + (P - Q - y)R + yQ = 0$, of discriminant

$$D(y) = (P - Q - y)^2 - 4yQ, \quad D(y) = 0 \iff y = (\sqrt{P} \pm \sqrt{Q})^2.$$

The two roots are exactly the spectrum edges in the variable $y = x^2$, and $D < 0$ strictly between them: the fixed point is complex precisely on the spectrum and real precisely off it, as it must be. Both recursions and the discriminant have been checked exactly.

Conjecture 10 for biregular G is therefore the statement that the asymmetric recursion, started from the leaves of the path tree, never produces $R = 0$ while $D > 0$. The natural invariant region is

$$R_A \leq -\alpha < 0, \quad R_B \in [1, 1 + Q/\alpha],$$

for a parameter $\alpha > 0$. A B -vertex with children in the region has $R_B = 1 + \sum 1/|R_A| \in (1, 1 + Q/\alpha]$, and a B -leaf carries $R_B = 1$, exactly the lower boundary. An A -vertex with its full complement of P children has $R_A \leq y - P\alpha/(\alpha + Q)$, so the region is closed precisely when $y \leq h(\alpha)$, where $h(\alpha) = \alpha(P - Q - \alpha)/(\alpha + Q)$. The following is elementary and is machine-checked in Lean 4 (`region_closure_bound`, `gap_edge_identity`):

$$(s - t)^2(\alpha + t^2) - \alpha(s^2 - t^2 - \alpha) = (\alpha - t(s - t))^2, \quad (6)$$

so $h(\alpha) \leq (s - t)^2$ for every α , with equality at $\alpha = t(s - t) = \sqrt{PQ} - Q$. The region therefore closes for every y strictly below the gap edge and for no y above it: the invariant region is calibrated exactly to the spectrum, and (6) is the reason.

Two structural facts explain the boundary between the proved and the open case.

The first is a reflection. The subdivision identity $\mu_{S(H)}(x) = x^{|E|-|V|} \mathcal{LM}(H, x^2)$ is due to Yan and Yeh [6], who also treat the regular and semiregular cases; see [4], Corollary 2.3, for a restatement through the multivariate matching polynomial. For H D -regular it reads $f_G(y) = \mu_H(y - D)$, and $\mu_H(-z) = (-1)^{|V(H)|} \mu_H(z)$, so

$$f_G(2c - y) = (-1)^p f_G(y), \quad c = (a - 1) + (b - 1), \quad (7)$$

and c is the midpoint of $[(s - t)^2, (s + t)^2]$ because $(s \pm t)^2 = c \pm 2\sqrt{m}$. The roots of f_G are therefore symmetric about the centre of the band, so for $b = 2$ the lower bound in (4) is *equivalent* to the upper one, and Theorem 11 is that equivalence. The reflection (7) fails as soon as both degrees exceed 2: it fails for $K_{3,4}$, $K_{3,5}$ and already for the regular $K_{3,3}$, so regularity does not rescue it. Since $b = 2$ holds exactly when G is the subdivision of an a -regular multigraph, (7) describes the whole of the phenomenon, and for $b \geq 3$ the lower bound is not a repackaging of the upper bound but carries information the upper bound does not. We note that (7) has no adjacency analogue: the companion identity $\varphi(S(H), x) = x^{|E|-|V|} \varphi(H, x^2 - D)$ holds as well, but $\varphi(H, \cdot)$ has a parity only for bipartite H , whereas μ_H always does. For the Petersen graph the matching-side values are symmetric about $D = 3$ and the adjacency-side values are not.

A consequence of the first is that the band edges are never roots, by an argument of a different kind.

Proposition 13. *Let G be (a, b) -biregular bipartite with $a, b \geq 2$, and suppose $m = (a - 1)(b - 1)$ is not a perfect square. Then neither $(s - t)^2$ nor $(s + t)^2$ is a root of f_G .*

Proof. f_G has integer coefficients and $(s \mp t)^2 = (a + b - 2) \mp 2\sqrt{m}$, so for m not a square the two band edges are conjugate over $\mathbb{Q}$; if one is a root of f_G , so is the other. Every root of μ_G is bounded by the spectral radius of a path tree $T_u(G)$ [9], and $T_u(G)$ is finite and embeds in the universal cover of G , the (a, b) -biregular tree $T_{a,b}$, with degrees bounded by those of $T_{a,b}$.

Now $\rho(T_{a,b}) = s + t$ is the supremum of ρ over finite subtrees of $T_{a,b}$, and that supremum is not attained: any finite subtree is a proper subgraph of a larger connected finite subtree, and ρ is strictly monotone under proper subgraph containment for finite connected graphs. Hence $\rho(T_u(G)) < s + t$, so $(s + t)^2$ is not a root of f_G , and therefore neither is $(s - t)^2$. $\square$

The hypothesis on m is needed for the argument, not merely for convenience: when m is a square the two band edges are rational and independent, and conjugacy says nothing. The smallest such case is $(a, b) = (5, 2)$, where $m = 4$ and the edges are 1 and 9. There the conclusion nevertheless holds, but for a different reason — the Heilmann–Lieb bound is strict for finite graphs, so Theorem 11 already excludes the lower edge. Proposition 13 is the statement for the cases those methods do not reach.

Evaluating f_G in $\mathbb{Z}[\sqrt{m}]$ confirms this and exhibits the conjugacy: for $K_{3,4}$, $K_{3,5}$, $K_{3,6}$, $K_{4,5}$ the two edge values are $53 \mp 30\sqrt{6}$, $72 \mp 40\sqrt{8}$, $91 \mp 50\sqrt{10}$, $1877 \mp 512\sqrt{12}$. For $b = 2$ the irrational parts vanish, as (7) requires. Proposition 13 is worth isolating because it uses no induction on vertex deletion, and so is not subject to the obstruction described next; it is the only tool we have found with that property, although it controls only the edges and not the interior of the gap.

The second point is that the lower bound, unlike the upper bound, is not monotone under vertex deletion. The Heilmann–Lieb argument for $|x| \leq 2\sqrt{\Delta - 1}$ inducts on vertex deletion, which is available because Δ only decreases. The quantity $|\sqrt{a - 1} - \sqrt{b - 1}|$ has no such monotonicity: deleting one vertex of the larger side of $K_{3,4}$ produces $K_{3,3}$, which is regular, and the gap collapses to zero. So the naive induction by vertex deletion, which is what yields the upper bound, cannot prove the lower one. We do not claim this rules out every inductive argument, only the direct analogue of the Heilmann–Lieb one. A lower bound of the same shape is known in the random setting: Brito, Dumitriu and Harris [7] show that a random bipartite biregular graph satisfies $\eta_{\min}^+ \geq \sqrt{d_1 - 1} - \sqrt{d_2 - 1} - \varepsilon_n$ with high probability. That is an asymptotic statement about adjacency eigenvalues of random graphs; Conjecture 10 is a deterministic statement about matching polynomial roots of every graph.

Two obstructions remain in the direct approach, both caused by cycles in G . First, a leaf of the path tree that is an A -vertex carries $R_A = y > 0$, of the wrong sign. This one is disposed of: a maximal self-avoiding walk ending at $w \in A$ contains all a neighbours of w , hence at least a vertices of B and therefore at least a vertices of A , so

$$p < a \implies \text{the path tree has no } A\text{-leaf,}$$

which covers every $K_{p,q}$ with $p < q$. Second, and this is what blocks the argument, an internal A -vertex of the path tree may have $k < P$ children, the walk having already consumed part of its neighbourhood; the bound $R_A \leq y - P\alpha/(\alpha + Q)$ then fails. With $k = 1$ closure would require $\alpha(1 - \alpha - Q)/(\alpha + Q) \geq y$, impossible for $b \geq 3$ since $Q \geq 2$. Deficiency occurs already in $K_{3,4}$, where the walk $a_1b_1a_2b_2a_3$ leaves a_3 with two children rather than three. What the argument as it stands establishes is that a *full* truncated biregular tree carries no eigenvalue in the gap, consistent with the deficient subtree P_8 carrying one; that is a statement about the tree and not about G . Controlling deficient A -vertices is the open problem.

The case $b \geq 3$ of the biregular family is the sharp open case, since $b = 2$ is Theorem 11. It has been tested at $d = 1$ on $K_{3,4}, K_{3,5}, K_{3,6}, K_{3,7}, K_{3,8}, K_{4,5}, K_{4,6}, K_{4,7}, K_{5,6}$ and a $(3, 2)$ design incidence graph, with smallest-root ratios between 1.51 and 3.33 and no root in a gap.

The natural attack is Godsil's recursion $R_v = x - \sum_u 1/R_u$, whose fixed points on the (a, b) -biregular tree are real precisely off the spectrum. Asking that the region $R_b \in [x, P/x]$, $P = a - 1$, $Q = b - 1$, be closed under the recursion reduces to $u^2 - (2P + Q)u + P^2 > 0$ at $u = x^2$, i.e. to $u < (2P + Q - \sqrt{Q^2 + 4PQ})/2$. This is satisfied throughout the gap with room to spare: the ratio of that threshold to $(\sqrt{P} - \sqrt{Q})^2$ ranges from 2.7 to 37.7 over the cases above. So the analytic part of an induction is not the difficulty.

What blocks it is the boundary. A leaf of the path tree carries $R = x > 0$, while the argument wants $R < 0$ at a -vertices, and mixed signs among the children of a b -vertex leave the sum uncontrolled. One might hope to exploit that the counterexample to the naive subtree argument, P_8 inside the $(3, 2)$ -biregular tree, is deficient at *internal* vertices, whereas a path tree is full at internal vertices; but that is false for graphs of small girth, where a self-avoiding walk can be blocked early — in K_4 the walk $0, 1, 2$ admits a single extension. We therefore leave $b \geq 3$ open.

A proof of Conjecture 10 cannot go through Floquet theory. For $b_1 = b$ the deck group of the universal cover is the free group F_b , and Floquet theory is exactly the statement that for an *abelian* deck group the fibres are indexed by the characters of that group; the two coincide only at $b = 1$, where $F_1 = \mathbb{Z}$. For $b \geq 2$ the correct index set is the unitary dual of F_b , and the relevant object is the adjacency operator in the reduced C^* -algebra of F_b . This is not merely a technical obstruction: the computations of Section 3 show that the per-component weight in the exponential formula, a single graph for $b = 1$, becomes a sum over the many non-isomorphic connected ℓ -covers for $b \geq 2$.

That framework is not empty. As $d \rightarrow \infty$, independent uniform random permutations are strongly asymptotically free [12], and consequently the spectrum of a random d -lift of G converges to $\text{spec}(T)$. Conjecture 10 is a finite- d statement about the expected characteristic polynomial rather than an asymptotic statement about typical lifts, and its content is precisely that the containment holds uniformly in d , with no exceptional set. What the $b_1 = 1$ case shows is that the interval in [2] is not the natural receptacle for the roots, and that the correct one is spectral rather than metric.

Related work. Two ingredients are classical and are used, not claimed. The factorization of the characteristic polynomial of a cyclic cover into twisted determinants at roots of unity is the block-circulant diagonalization [14, 15]; in the infinite-cover limit it is the Floquet–Bloch decomposition of a periodic discrete operator, and $y = -P/(2Q)$ is the corresponding Floquet discriminant. The identity $\Phi_{G,r} = \chi_G \cdot \mu_{r-1,G}$ is [2]. Closest in form is McSorley and Feinsilver [13]. For a path whose edges are cyclically labelled with t labels they establish a (τ, Δ) -recurrence $\Phi_N = \tau\Phi_{N-1} - \Delta\Phi_{N-2}$, which is the algebraic skeleton of (3), and their Theorem 4.4 solves it as $\Delta^{N/2}U_N(\tau/2\sqrt{\Delta})$, exactly the form used here. In the univariate specialization their τ is the matching polynomial of the cycle C_t and $\Delta = (-1)^t x_1 \cdots x_t$ is a monomial; passing between their counting normalization and the characteristic one used here is the homogenization $V_d \mapsto c^d V_d$, under which a monomial Δ becomes $\mu_{G-V(C)} = 1$. Their Theorems 4.4 and 4.5 together give $U_{dn+n-1} = U_d(T_n)U_{n-1}$, the identity invoked in Section 1 to pass between Theorem 1 and Hall's conjecture, so that bridge was already available in 2009. Their parameter is the number of label periods along a path and their graphs are paths, cycles and rooted trees; there is no covering and no cycle with trees attached. What is added here is the pair $(\mu_G, \mu_{G-V(C)})$, whose second entry is a genuine polynomial recording

the attached trees rather than a monomial, together with the covering interpretation that makes the recurrence a statement about $\mu_{d,G}$.

The boundary is sharp. For $b_1(G) = b$ the lift is determined by b permutations, that is by an action of the free group F_b on $[r]$, whose orbits are the components of the lift. The exponential formula still applies, in the form $N_r = \sum_{\ell} \binom{r-1}{\ell-1} C_{\ell} N_{r-\ell}$ with $N_r = (r!)^b \Phi_{G,r}$ and C_{ℓ} the sum of χ over the transitive b -tuples on $[\ell]$; we have verified this for $b \leq 3$. What fails is the atom. For $b = 1$ every transitive tuple gives the *same* cover G_{ℓ} , so $C_{\ell} = (\ell - 1)! \chi_{G_{\ell}}$: one graph per ℓ , with $\chi_{G_{\ell}}$ Chebyshev in ℓ . For $b \geq 2$ the transitive tuples number $(\ell - 1)!$ times the count of index- ℓ subgroups of F_b and spread over many non-isomorphic covers with distinct spectra, so C_{ℓ} carries no such structure. Accordingly Theorem 6 is not merely unproved for $b \geq 2$: it is false there already at $d = 1$, as we checked for the theta graph, the bowtie, K_4 , $K_{2,3}$ and two triangles joined by an edge.

Theorem 6 was verified by exact integer computation against the definition of $\mu_{d,G}$ as an average of matching polynomials over all d -covers, and against brute-forced lift averages, on thirteen named unicyclic graphs for $r \leq 6$ and on a random sample of thirty-four for $r \leq 5$; Lemma 7 was checked on eleven.

4 The formalization

The core algebraic derivation — the exponential formula, the closed form of Theorem 1 for all n, r (equivalently, everything from the cycle weights (1) to the closed form), its factorization, and the parity Corollary 5 — is machine-checked in Lean 4 over Mathlib (v4.30). The main entry points:

- `Paper2.expFormula` — the exponential formula (2) in integer form $r A_r = \sum_k r^{(k)} f(k) A_{r-k}$ over any commutative ring, proved via Mathlib’s Cauchy cycle-type count (`Equiv.Perm.card_of_cycleType`);
- `Paper2.thm1_Sr` — Theorem 1 in the form: the expected cycle weight (1) equals the closed form, for all n, r , over any $\mathbb{Q}$ -algebra domain;
- `Paper2.cG_factored`, `Paper2.quotient_parity` — the factorization and Corollary 5.

The three remaining corollaries are formalized only in part: Corollary 2’s generating function appears as the finite recurrence (2) rather than a literal power-series identity; of Corollary 3 only the underlying trigonometric factorization (`cor1_factored`) is checked, not the “all roots in $[-2, 2]$ ” statement end-to-end; and Corollary 4 (`cor2_phi3`) is a bounded `native_decide` over $n, r \leq 6$.

Trust base for the general theorems: `propext`, `Classical.choice`, `Quot.sound` only (checked with `#print axioms`); the bounded `native_decide` checks additionally invoke `Lean.ofReduceBool`. No sorry. The single remaining unformalized ingredient of the *graph-theoretic* statement is the sentence before (1): a lift of a cycle is a disjoint union of cycles — elementary covering-space combinatorics for which Mathlib currently lacks the vocabulary.

Remark 14. Everything here is stated for the uniform permutation model; by [2] the quotient coincides with the $(r - 1)$ -matching polynomial, so Theorem 1 can alternatively be deduced from [3]. We believe the generating function of Corollary 2, the evaluation of Corollary 4, and the formalization are new.

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